

ZAIDAN DZAKI PRASETYA ADI, S.T.

Surabaya, East Java, Indonesia | +62 812-5255-8679 | zaidandzakipa34@gmail.com
[linkedin.com/in/zaidaaandpa](https://www.linkedin.com/in/zaidaaandpa) | zaidandzaki.github.io/zaidandzakipa34

Professional Summary

Detail-oriented Civil Engineering graduate from Institut Teknologi Sepuluh Nopember (ITS, GPA 3.74/4.00) specializing in **Structural Analysis and Building Information Modeling (BIM)**. Proven track record in finite element modeling and code-compliant design of complex bridge systems (cable-stayed, tied-arch, steel truss up to 240 m span) and multi-storey reinforced concrete structures per SNI standards. Highly skilled in leveraging **Autodesk Revit, SAP2000, ETABS, and GEO5** for multi-disciplinary coordination, clash detection, and geotechnical analysis, seamlessly bridging theoretical design with constructability and field execution.

Core Competencies

- Structural Engineering & Analysis:** Finite Element Modeling (FEM), Seismic Design (Equivalent Static & Response Spectrum per SNI 1726:2019), RC Detailing (SNI 2847:2019), Steel Design (AISC/SNI 1729), Cable Dynamics, Wind Load Analysis
- BIM & Technical Drafting:** Autodesk Revit (Structural & Architectural LOD 300–350), Navisworks (Clash Detection), AutoCAD, Civil 3D, Bar Bending Schedules (BBS), Quantity Take-Off (QTO / BOQ)
- Geotechnical & Foundations:** GEO5 (Retaining Walls, Slope Stability, Deep/Piled Foundations), Soil Laboratory Testing, CPT/Borehole Soil Investigation
- Software & Computational Tools:** SAP2000, CSI ETABS, MIDAS Civil, IDEA StatiCa, SketchUp, Lumion, SMath Studio, MATLAB, QGIS, MS Project, MS Excel (Advanced Engineering Worksheets)

Education

- Institut Teknologi Sepuluh Nopember (ITS)** **Surabaya, Indonesia**
Bachelor of Engineering (S.T.) in Civil Engineering – GPA: 3.74 / 4.00 *Aug 2022 – Jan 2026*
- **Final Capstone Project:** *Structural Redesign and Modification of the 240 m Aik Bari IV Cable-Stayed Bridge* – Engineered a continuous orthotropic steel box girder system with semi-harp stay cables, conducting balanced cantilever construction staging, non-linear cable pretensioning, and 3D seismic response spectrum evaluations to ensure full compliance with Indonesian Bridge Design Codes (SNI 2833 / RSNI T-02).
- SMAN 1 Surabaya** **Surabaya, Indonesia**
Natural Sciences and Mathematics – Average Score: 90.15 / 100 *Jul 2019 – Jun 2022*

Professional Experience

- Independent BIM Modeler & Consultant** **Surabaya, Indonesia**
Freelance Structural & BIM Engineer *May 2024 – Feb 2026*
- **Engineered parametric 3D BIM models** (LOD 300–350) for complex bridge systems (steel truss, tied-arch, 240 m cable-stayed) and a 10-storey commercial building using **Autodesk Revit**.
 - **Integrated multidisciplinary models** (superstructure, substructure, and MEP systems) into Navisworks, identifying and resolving over 50+ geometric clashes prior to construction tender.
 - **Extracted dynamic 2D construction drawings** and automated structural Quantity Take-Off (QTO), yielding 98% accuracy against manual BoQ calculations and mitigating potential material wastage.
- Building Assessment Surveyor** **Surabaya, Indonesia**
Structural Condition Assessment Team (Contract) *Jul 2026 – Aug 2026*
- **Conducted forensic structural condition surveys** across multiple institutional facilities, identifying crack patterns, concrete spalling, and rebar corrosion on primary structural load-bearing frames.
 - **Engineered technical assessment matrices** and repair recommendations for municipal stakeholders, coordinating field inspections and ensuring regulatory compliance with zero operational downtime.
- Construction of 8-Storey Muhammadiyah Babat Hospital** **Lamongan, Indonesia**
Civil Engineering Site & BIM Intern *Dec 2024 – Feb 2025*
- **Supervised structural execution and QA/QC** for an 8-storey reinforced concrete hospital, performing slump tests, cylinder compressive strength verification, and rebar tolerance inspections.
 - **Synthesized automated Bar Bending Schedules (BBS)** using Autodesk Revit, reducing rebar cutting scrap by approximately 7% and optimizing steel placement logistics on site.
 - **Produced daily/weekly construction progress tracking** and heavy equipment cycle productivity reports, supporting critical-path scheduling adjustments to prevent project milestones slippage.
- Municipal Infrastructure Technical Audit** **Lamongan, Indonesia**
Forensic Audit Assistant / Expert Team Member *Apr 2025 – Jun 2025*
- **Audited physical construction discrepancies** for public municipal infrastructure, performing reverse structural volume calculations and reconciling field conditions against Approved-for-Construction (AFC) drawings.
 - **Authored comprehensive technical forensic reports** adopted by municipal authorities for legal and contractual accountability.

PT Geomarine Survey Dinamika <i>Geotechnical Engineering Intern</i> • Supervised deep soil exploration including Cone Penetration Testing (CPT) and mechanical core drilling for high-voltage PLN transmission tower foundations in Lamongan. • Executed geotechnical laboratory analyses (sieve analysis, Atterberg limits, triaxial, moisture content, specific gravity) for coastal infrastructure projects (PLTMG Sumbawa, PLTMG Bima, and Barito River Jetty).	Surabaya, Indonesia <i>Jul 2024 – Aug 2024</i>
PT Tunas Jaya Sanur <i>Site Supervision Intern</i> • Supervised architectural and MEP finishing works, verifying elevation levels, plumbness, and dimensional accuracy across exterior drainage and interior structural elements against design specifications.	Surabaya, Indonesia <i>Jan 2024 – Feb 2024</i>

Engineering Competitions & Structural Design Awards

IDEERS 2025 – International Earthquake-Resistant Structure Competition <i>8th Place & Innovation Award</i>	Oct 2025
• Designed and fabricated a 60 cm seismic-resistant frame building model integrated with custom elastomeric base dampers; successfully withstood simulated seismic acceleration forces exceeding 800 Gal on the shake table out of 49 international university teams.	
National Bridge Design Competition CIVEX 2024 <i>1st Place Winner & Favorite Award</i>	Nov 2024
• Spearheaded structural finite element analysis and aesthetic concept for a 200 m pedestrian cable/arch bridge, recognized for optimized deck-stress distribution, aerodynamic performance, and inclusive public space.	
National Bridge Competition CIVEST 2025 <i>2nd Runner-Up</i>	Jun 2025
• Engineered “Rajansa Bridge,” a double-span tied-arch steel bridge (2 × 80 m) utilizing MIDAS Civil and SAP2000, optimizing hanger cable configurations for maximum stiffness and accelerated constructability in Aceh.	
National Bridge Competitions (CIA 2024 & 8th CREATION) <i>2nd & 3rd Winner</i>	2023 – 2024
• Secured multiple national podiums for designing 60 m to 120 m steel truss/arch bridges, balancing minimal material weight ratios with maximum live-load capacity under stringent deflection constraints.	

Certifications & Extracurricular Leadership

Structural Analysis Training: Special RC Moment Frames (SRPMK) with CSI ETABS	May 2026 – May 2027
• Completed intensive 40+ hour training on 3D high-rise building seismic analysis, response spectrum derivation, dual-system design, and capacity-based reinforced concrete beam-column joint detailing per SNI 1726:2019 / SNI 2847:2019.	
ITS Technical Crew – Indonesian Bridge Competition (KJI 2025)	Sep 2025 – Oct 2025
• Coordinated material fabrication and structural testing jigs for a 5-meter steel truss bridge model; calibrated precision LVDT/dial gauges to record elastic deflections during static proof loading.	